

Exponential function

Here a is nonzero, b is positive, b is not 1, and x is real.

$$f(x) = ab^x$$

Why must the base be positive?

Variable in the exponent

$$3^x \quad x^3$$

Which expression is exponential, and which is a power function?

A forbidden base

$$(-4)^x$$

Would this expression give a real output for every real x ?

A base of one

$$1^x$$

Why does this function fail to describe exponential growth or decay?

Zero and negative exponents

$$f(x) = 6 \cdot 2^x$$

What are $f(0)$ and $f(-2)$?

Fractional input

$$f(x) = 4 \cdot 9^x$$

What is the exact value of $f(1/2)$?

Order of operations

$$f(x) = 3 \cdot 2^x + 1$$

What is $f(2)$?

Equal input steps

$$f(x) = 5 \cdot 3^x$$

By what factor does the output change when x increases by 1?

Growth factor

P is a population count and t is years after the initial measurement, $t \geq 0$.

$$P(t) = 800(1.06)^t$$

What annual percent change does the model represent?

Decay factor

V is equipment value in dollars and t is years since purchase, $t \geq 0$.

$$V(t) = 12000(0.85)^t$$

What annual percent decrease does the model represent?

Addition or multiplication

Both outputs are amounts in dollars; t is elapsed years, $t \geq 0$.

$$L(t) = 40 + 8t \quad E(t) = 40(1.2)^t$$

Which model increases by a constant amount, and which by a constant percentage?

Writing a growth model

An account starts at \$600 and increases by 4% each year. Assume no deposits or withdrawals and $t \geq 0$.

What exponential model gives the account balance after t years?

Writing a decay model

A sample begins with 90 grams and loses 12% of its remaining mass each hour. Assume the same fractional decrease and $t \geq 0$.

What exponential model gives the remaining mass after t hours?

Two points with an initial value

Assume $a > 0$ and $b > 0, b \neq 1$.

$$f(x) = ab^x$$

What exponential function passes through (0, 5) and (2, 45)?

Two points without an initial value

Assume $a > 0$ and $b > 0, b \neq 1$.

$$f(x) = ab^x$$

What exponential function passes through (1, 12) and (3, 48)?

Compound interest

P is principal, r is the annual interest rate as a decimal, n is compoundings per year, and t is elapsed years.

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

What does n count in this formula?

Monthly compounding

Deposit \$1,500 at 4.8% annual interest compounded monthly, with no deposits, withdrawals, or fees.

What is the balance after 3 years, rounded to the nearest cent?

A quarterly growth factor

An account earns 6% annual interest compounded quarterly, with no deposits or withdrawals.

What factor multiplies the balance each quarter?

The natural base

The number e is the base of the natural exponential function.

$$e \approx 2.71828$$

Is e raised to a real power always positive?

Evaluating a natural exponential

$$f(x) = 7e^{-0.4x}$$

What is $f(3)$, rounded to four decimal places?

Continuous compounding

P is principal, r is the annual interest rate as a decimal, and t is elapsed years.

$$A = Pe^{rt}$$

How does t differ from the number of compounding periods?

A continuously compounded balance

Deposit \$2,000 at 3.5% annual interest compounded continuously, with no deposits, withdrawals, or fees.

What is the balance after 5 years, rounded to the nearest cent?